Want to solve homogeneous matrix equation:

$$\underset{m \times n}{A}\ \underset{n \times 1}{x} = \underset{m \times 1}{0}$$

Again keep in mind, these are not real numbers they are matrices. Often will use this notation without directly clarifying that. Important that you be wary of context.

How to solve this? Write the coefficient matrix A in REF. Slighty different than earlier when we looked at augmented matrix - why?

This being homogeneous, in augmented matrix the last column is all 0s. And naturally stays that way even if we apply elementary row operations. So that being the case, system certainly is consistent, no pivot in final column. But really given that column stays all 0s, little point writing that at each juncture, is understood.

This is one reason why homogeneous is much more convenient than nonhomogeneous, one less column to worry about when doing row operations. And if matrix is large that can be significant.

1.5.9 Solve $Ax = 0$ for

$$A = \begin{bmatrix} 3 & -9 & 6 \\ -1 & 3 & -2 \end{bmatrix}$$

Replace R2 by 3R2 + R1. Note that this is technically two elementary row operations since we multiply that row by 3 then add the other, replacing R2 by 3R1 + R2 would be a single row operation in comparison.

$$\sim \begin{bmatrix} 3 & -9 & 6 \\ 0 & 0 & 0 \end{bmatrix}$$

Nonpivot columns circled so x_2, x_3 free.

Only one relevant equation:

$$3x_1 - 9x_2 + 6x_3 = 0$$

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$$x_1 - 3x_2 + 2x_3 = 0$$

Thus

$$x_1 = 3x_2 - 2x_3$$

since latter two are free, this is answer. Express as

$$\begin{bmatrix} 3x_2 - 2x_3 \\ x_2 \\ x_3 \end{bmatrix} = \underbrace{x_2 \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}}_{\text{parametric vector form}}$$

Latter form often used. Nice thing about it is that it tells us that the set of vectors satisfying this equation is <u>spanned</u> by the two vectors:

$$\left\{ \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Hence one reason why concept of span is useful.

How to solve/express answer to nonhomogeneous equation $Ax = b$? Two possible methods:

(1) Set up augmented matrix $[A \mid b]$ and solve as usual.
(2) Find one <u>particular solution</u> to the nonhomogeneous equation and add that to the general solution to the homogeneous equation. Same idea that we used when solving second order differential equations with constant coefficients while finding particular solution via undetermined coefficients or variation of parameters.

Will again see those ideas later in this course!

Why does (2) work? Say $Ax = b$ has some solution y. Then for another prospective solution z we overall have:

$$Ay = Az = b$$

Meaning

$$Az - Ay = 0$$
$$A(z - y) = 0$$

Thus $z - y$ is a solution to the homogeneous equation $Ax = 0$. So rephrasing

$$z = y + (z - y)$$

That is, the new prospective solution can be written as a sum of the original solution y and a solution to the homogeneous equation.

Thus any solution to the nonhomogeneous equation can be written as that particular solution plus a a solution to the homogeneous equation, as desired.

Hence the general solution to the nonhomogeneous is that particular solution plus the general solution to the homogeneous equation.

<u>Ex:</u> Fully solve Ax = b for

$$A = \begin{bmatrix} 3 & -9 & 6 \\ -1 & 3 & -2 \end{bmatrix}, \quad b = \begin{bmatrix} -12 \\ 4 \end{bmatrix}$$

Using first method, augmented matrix is:

$$\begin{bmatrix} 3 & -9 & 6 & -12 \\ -1 & 3 & -2 & 4 \end{bmatrix}$$

Using same row operations as in homogeneous case, this is row equivalent to:

$$\sim \begin{bmatrix} 3 & -9 & 6 & -12 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Again only one relevant equation which is:

$$3x_1 - 9x_2 + 6x_3 = -12$$

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$$x_1 - 3x_2 + 2x_3 = -4$$

Since x_2, x_3 still free, we solve for x_1 in terms of them:

$$x_1 = -4 + 3x_2 - 2x_3$$

General solution is thus:

$$\begin{bmatrix} -4 + 3x_2 - 2x_3 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

What if we tried to solve with second method? Well homogeneous solution we have, just need particular solution. And since only one relevant equation, namely

$$x_1 - 3x_2 + 2x_3 = -4$$

can just find a solution via plugging in (ok if more equations, not so simple, might have to eliminate variables). But here say taking first two to be 1, gives

$$\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

So different looking particular solution, but same homogeneous part as should be (ok could take constant multiple of those but essentially the same).

Either way similar to homogeneous solution, just that additional vector in front. But that makes a very big difference! Why?

The two vectors on the right of the solution are the ones which span the solution of the homogeneous equation. But which two span the nonhomogeneous solution?

Geometrically seems like solution set for nonhomogeneous is same type of thing as homogeneous. And in a sense it is, but more to that than you might think.

A big difference is the homogeneous solution always passes through origin while nonhomogeneous never does. Similar to how in a homogeneous differential equation, 0 function always a solution, but it never is in nonhomogeneous.

And the implication of that is broader than you might initially think. We asked what vectors spanned the nonhomogeneous solution. But doesn't the span of any set of vectors always contain the origin via taking every scalar multiple to be 0?

It does! So the nonhomogeneous solution is not the span of a set of vectors like the homogeneous solution, big difference!

1.7 Linear Independence

Again an m x n matrix can be thought of as n vectors in $\mathbb{R}^m$ via looking at the n columns.

If $n \geq m$, is it likely these vectors span the full $\mathbb{R}^m$? Seems likely, especially if n is much larger.

What if $m > n$ though? Is it even possible for them to span $\mathbb{R}^m$? We think not but to that end, if not how "big" a piece do they create in $\mathbb{R}^m$?

Say in $\mathbb{R}^{100}$ we have 50 vectors. Do they create something which is $\mathbb{R}^{50}$? Not exactly $\mathbb{R}^{50}$ perhaps since vectors have 100 components but maybe something similar? E.g. a plane in $\mathbb{R}^3$ is similar to $\mathbb{R}^2$ but not identical.

Maybe, but what if one vector is a multiple of another rendering one of them redundant? Then instead something like $\mathbb{R}^{49}$? But what if more redundancy and overall is more like $\mathbb{R}^{40}$? Well how can you assess that?

To that end, the concept of a set of vectors being <u>linearly independent</u> is very important in many contexts.

A set of vectors $\{ v_1, v_2, \ldots, v_n \}$ is linearly independent if the system

$$c_1 v_1 + c_2 v_2 + \ldots + c_n v_n = 0$$

has only the <u>trivial</u> solution: $c_1 = c_2 = \ldots = c_n = 0$

Linearly dependent naturally means this does have a nontrivial solution.

How do we check that in $\mathbb{R}^m$?

As we did earlier, catalog these vectors as the columns of an $m \times n$ matrix as such:

$$A = \begin{array}{c} \begin{bmatrix} \boxed{\begin{matrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{matrix}} & \boxed{\begin{matrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{matrix}} & \cdots & \boxed{\begin{matrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{matrix}} \end{bmatrix} \\ \begin{matrix} v_1 & v_2 & & v_n \end{matrix} \end{array}$$

And again, as we showed earlier, a linear combination of these columns/vectors being 0 is equivalent to solving the matrix equation $Ax = 0$.

Hence why this ties in nicely with the prior section which in part dealt with solving homogeneous equations of that sort.

So how might knowing about linear independence help us? Well if we have a set of vectors, and we are curious the "size" of the span, checking if the set is linearly independent or dependent could be a good start.

If it's the latter, i.e. there is redundancy, maybe we can throw a vector out of the set and maintain the same span (and possibly more later if it stays dependent).

If so, can we pick which one to throw out? Naturally we might prefer to throw out a vector with many nonzero entries compared to one with many since the former is harder to work with. For example

$$\left\{ \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

This set is linearly dependent since last vector is the sum of the prior two.

So we can throw out any of last three and maintain the same span. But we'd like to throw out the first since it's the most annoying to work with. We can't though as it is not a linear combination of other three as they all have 0 in the top coordinate while the first doesn't.

That is, in a linearly dependent set of vectors, one vector in the set is a linear combination of the others, but that doesn't mean every vector in the set is.

And that makes our job harder as even if we know how "redundant" a set of vectors is (i.e. how many can can be thrown out while keeping span the same), we have to know which ones can be thrown out, that can't be done arbitrarily.

So are some issues here. Say we have fifty vectors, but we know we only need forty to create the same span.

Ok, figuring out we need exactly forty is one thing that has to be generalized, how can you determine that?

But assuming we know that, i.e. we can throw out ten vectors, how do we do that?

Say we figure out one specific vector is redundant by showing it's a linear combination of the others so we throw it out. But what about the other nine?

Say another one was originally redundant too (perhaps was a multiple of the one we threw out) does it stay redundant?

It may not! Being a multiple of the one we threw out would definitely make that possible!

Given that, are we forced to reexamine the set of forty nine vectors left to see which are redundant now?

We certainly hope not as that would be very annoying to have to do that at every step!

There are many dilemmas like the above that we will face as we learn more about these ideas.